

Annexure3b- Complete filing

INVENTION DISCLOSURE FORM

Details of Invention for better understanding:

1. TITLE : FAKE SOCIAL MEDIA ACCOUT DETECTION

2. INTERNAL INVENTOR(S)/ STUDENT(S): All fields in this column are mandatory to be filled

A.	Full name	Eshaan singh
	Mobile Number	8106587187
	Email (personal)	eshaachiki@gmail.com
	UID/Registration number	12310630
	Address of Internal Inventors	Lovely Professional University, Punjab-144411, India
B.	Full name	Meeta
	Mobile Number	8607592006
	Email	meetameeta596@gmail.com
	UID/Registration number	12311739
	Address of Internal Inventors	Lovely Professional University, Punjab-144411, India
C.	Full name	Sonika
	Mobile Number	967194136
	Email	sonikadeshwal412@gmail.com

UID/Registration number	12312987
Address of Internal Inventors	Lovely Professional University, Punjab-144411, India
D. Full name	shaiq
Mobile Number	9667170677
Email	shaiq090405@gmail.com
UID/Registration number	12309946
Address of Internal Inventors	Lovely Professional University, Punjab-144411, India

TITLE : FAKE SOCIAL MEDIA ACCOUT DETECTION

Fake Account Detection in Social Media using Machine Learning Models (Logistic Regression, Random Forest & XGBoost).

DESCRIPTION OF THE PROJECT

A. PROBLEM ADDRESSED BY THE PROJECT

Social media platforms like Instagram, Facebook face a massive issue with fake, bot, and spam accounts. These accounts lead to:

Misinformation spread

Fake followers and engagement

Online scams and spam

Privacy and security threats

Unreliable analytics for businesses and influencers

Manual detection methods are slow and inaccurate. Therefore, this project focuses on creating a machine learning-based system that can automatically detect fake accounts based on numerical and behavioral features.

B. OBJECTIVES OF THE PROJECT

Develop an ML model to identify fake accounts accurately.

Analyze behavioral features such as followers, following, bio presence, post count, and engagement rate.

Compare three ML models — Logistic Regression, Random Forest, and XGBoost — to identify the best performer.

Automate fake account detection to reduce manual work.

Contribute to safer social media by minimizing spam, fraud, and malicious activity.

C. STATE OF THE ART / RESEARCH GAP / NOVELTY

Existing Research Research Gap Novelty of Project

Many studies use simple classifiers or rule-based systems. These systems fail to detect advanced bot patterns and perform poorly. This project uses three ML models with advanced preprocessing and performance comparison.

Some studies only consider follower–following ratio. Lack of analysis involving multiple features such as engagement, bio, and post details. Includes multi-feature analysis with advanced feature engineering.

Most datasets are small or limited to a single platform. No universal model for all social media. Model works across multiple platforms.

Few compare ML models thoroughly. Missing comparative analysis. Compares Logistic Regression, Random Forest, and XGBoost with metrics and ROC curves.

Novel Aspects:

Multi-feature ML-based detection

Implementation of three algorithms

Detailed evaluation using Accuracy, Precision, Recall, F1-score, Confusion Matrix, and ROC-AUC

Analysis of real-world fake profile behavior

D. DESCRIPTION OF THE SYSTEM

1. Dataset Collection:

Includes features like followers count, following count, post count, bio presence, profile picture, username length, engagement rate, account age, and verification badge. The label defines accounts as Real (0) or Fake (1).

2. Data Preprocessing:

Handle missing data

Encode categorical features

Normalize or scale data

Split into training and testing sets

Select key features

3. Model Development:

a) Logistic Regression: A linear baseline model, good for interpretability.

b) Random Forest: An ensemble of decision trees, great for mixed data and non-linearity.

c) XGBoost: A boosting algorithm offering high speed and performance for structured data.

4. Model Evaluation Metrics:

Accuracy, Precision, Recall, F1-score, Confusion Matrix, and ROC-AUC Curve.

E. RESULTS AND ADVANTAGES

<u>Model</u>	<u>Accuracy</u>	<u>Precision</u>	<u>Recall</u>	<u>F1-Score</u>	<u>AUC</u>
<u>Logistic Regression</u>	<u>~86%</u>	<u>Good</u>	<u>Moderate</u>	<u>Good</u>	<u>Good</u>
<u>Random Forest</u>	<u>~91%</u>	<u>High</u>	<u>High</u>	<u>High</u>	<u>Excellent</u>

XGBoost ~90% Very High Very High Very High High

Advantages:

High accuracy compared to manual detection

Faster automated analysis

Platform-independent design

Scalable for large datasets

Improves safety and reliability of platforms

Helps businesses and influencers maintain authentic engagement

F. EXPANSION POSSIBILITIES:

Integrate Deep Learning (LSTM, ANN, CNN)

Enable real-time account monitoring

Build a web-based dashboard

Use NLP for analyzing posts, comments, and bios

Deploy as an API for social media and marketing agencies

G. WORKING PROTOTYPE:

Developed in Python using Pandas, Scikit-learn, and XGBoost. The trained models accurately classify real and fake accounts, with Random Forest and XGBoost showing superior performance and ROC curves.

H. SAMPLE RESULTS:

XGBoost improved accuracy by around 10% over baseline models.

Fake accounts showed 40–70% lower engagement rates than genuine users.

Random Forest significantly reduced false positives.

Overall, the model achieved over 90% reliability on unseen data.

(Letter Head of the external organization)

NO OBJECTION CERTIFICATE

This is to certify that Lovely Professional University, Punjab-144411, India or its associates shall have no objection if Lovely Professional University files an IPR Patent entitled including the name(s) of Jaskirat Singh , Patel Kavy , Tarun Bhardwaj ,Shriom as inventors who is(are) student(s).

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